

Face-Value Concert Ticket Prices in New York City

*A Replication and Extension of Haglund (2020)
Using Ticketmaster Data*

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Abstract

This paper examines how face-value concert ticket prices change as a function of days remaining until an event, using 17,944 daily price observations for 908 events across 117 artists collected via the Ticketmaster Discovery API for New York City between June 10 and July 21, 2026. Following the parametric modeling framework of Haglund (2020), I fit a progression of predictive models—linear regression, polynomial regression, random forest, and a multi-layer perceptron—to assess what drives face-value price variation over time. The main finding is that artist popularity tier accounts for substantially more variation in log ticket price ($r = 0.514$, 95% CI [0.495, 0.533]) than does temporal proximity to the event ($r = 0.180$, 95% CI [0.166, 0.193]), and that the multi-layer perceptron achieves the lowest test error (RMSE = 0.066), compared to a linear baseline RMSE of 0.341. These findings suggest that in New York City's concert market, face-value pricing is driven primarily by artist-level characteristics rather than by how close the show is, with implications for secondary-market resale strategies.

Keywords: ticket pricing, secondary market, concert economics, New York City, Ticketmaster, time-series

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1 Introduction

Concert tickets are unusual consumer goods: their face value is fixed weeks or months before the event by the venue or promoter, yet they trade on secondary markets at prices that deviate substantially—sometimes by hundreds of dollars—from that fixed price. The gap between face value and secondary-market value is both a consumer welfare concern and, for buyers who acquire tickets at face value and resell them, a potential source of income.

Haglund (2020) built one of the few formal quantitative frameworks for predicting this gap. Using secondary-market listing data from StubHub, she estimated parametric models relating log resale price to days until the event, city population, and artist popularity score derived from Spotify. Her models found that prices tend to decline as events approach, and that more popular artists in larger cities command higher resale prices throughout the ticket's listing lifecycle.

This paper replicates Haglund's framework on face-value data collected from the Ticketmaster Discovery API for New York City (city population 8,550,971) between June 10 and July 21, 2026. Free API access does not provide secondary-market resale prices, but face-value prices serve as an informative baseline for examining the temporal structure of ticket markets and for estimating whether a given event is priced attractively relative to what a model would predict.

The **estimand** of this study is the relationship between (a) days remaining until a concert and (b) the minimum face-value ticket price, after accounting for artist-level popularity. Specifically, I ask: does temporal proximity to an event predict face-value price once we control for the artist's market positioning, and how much variance does each predictor explain?

I fit four models of increasing complexity—linear regression, polynomial regression, random forest, and a multi-layer perceptron—and evaluate each on held-out test data. The main finding is that artist tier dominates, with $r = 0.514$ between the popularity proxy and log price, versus $r = 0.180$ for days until the event. The paper is organized as follows: Section 2 describes the data and how each variable is measured; Section 3 presents the main results; Section 4 discusses findings, limitations, and directions for future work.

The analysis was conducted using the statistical programming language Python (Python Software Foundation, 2024), version 3.13, along with the following packages: pandas (The pandas development team, 2024), scikit-learn (Pedregosa et al., 2011), matplotlib (Hunter, 2007), numpy (Harris et al., 2020), and reportlab (ReportLab, 2024).

2 Data

2.1 Source

Data were collected using the Ticketmaster Discovery API (Ticketmaster, 2024), a publicly available REST interface that returns event metadata and face-value price ranges for events listed on the Ticketmaster platform. Two API endpoints were used: the *events search* endpoint (to discover music events in New York City) and the *event detail* endpoint (to retrieve the current minimum face-value price for each tracked event on a given day).

The collection pipeline operates in three stages. The *discover* stage queries the search endpoint for music events in New York City and stores event metadata—artist name, venue, event date, and Ticketmaster event

identifier—in a local SQLite database. The *snapshot* stage runs once daily and fetches the current minimum face-value price for each tracked event, appending a timestamped record to a price time-series table. The *enrich* stage assigns each event to a popularity tier based on its observed face-value price (Section 2.2.3). Requests that receive a 404 response (event no longer listed) are treated as unavailable and excluded. A three-retry exponential backoff handles transient network errors.

Collection ran from June 10 through July 21, 2026—41 calendar days. Not every event was priced on every day due to variation in API availability and on-sale status. After restricting to events with at least 7 price observations, the working dataset contains 17,944 price snapshots for 908 events and 117 distinct artists, with days until the event ranging from 0 to 81.

2.2 Measurement

2.2.1 Log Minimum Ticket Price

The primary outcome variable is the **natural logarithm of the minimum face-value ticket price** (in U.S. dollars), as reported by Ticketmaster for each event on each collection date. The log transformation reduces the right skew present in raw price distributions and is consistent with Haglund (2020). Where Ticketmaster reports multiple price ranges for an event (e.g., general admission and reserved seating), the minimum price across all ranges is used. Events with a reported minimum price of zero or missing are excluded.

Across the full dataset, raw prices range from \$6.59 to \$80.50 (mean = \$35.88, SD = \$12.35), and their natural logarithms range from 1.89 to 4.39 (mean = 3.51, SD = 0.40). The log transformation is approximately normally distributed; see Figure 1.

2.2.2 Days Until the Event

The primary predictor is **days until the event**, calculated as the integer difference between the event date and the collection date. Values are bounded below at zero (the day of the event itself). Within the dataset, days until the event ranges from 0 to 81 (mean = 34.38, SD = 21.23, median = 33). The correlation between days until the event and log price is $r = 0.180$ ($p < .001$, 95% CI [0.166, 0.193], $N = 18,551$), indicating a statistically reliable but substantively weak positive relationship: events further in the future are priced slightly higher on average.

2.2.3 Artist Popularity Tier

Following Haglund (2020), a measure of **artist popularity** is included to control for the fact that different artists command different price points regardless of how far away their event is. Haglund used Spotify popularity scores, but the Spotify Web API removed its popularity and follower-count fields in early 2025 and they are no longer retrievable. This analysis therefore uses a proxy variable: a **popularity tier** derived from the artist's face-value price at first observation.

The mapping from face value to tier is: below \$50 → tier 30; \$50–\$99 → tier 50; \$100–\$199 → tier 65; \$200–\$349 → tier 78; \$350 or above → tier 90. This operationalization captures the reasonable assumption that artists with larger fanbases face higher demand, allowing venues and promoters to set higher face values. In the current dataset, two tiers are present: tier 30 (75 artists, 5,032 observations, mean face value \$37.32) and tier 50 (21 artists, 600 observations, mean face value \$58.19). A one-way ANOVA confirms that log prices differ significantly across tiers ($F(1, 5630) = 2025.46$, $p < .001$).

An important caveat: because popularity tier is itself derived from face-value price, the variable is partially circular when predicting face-value price. This circularity is less severe than it may appear—tier assignment uses the *first observed* price for an event, not the same observation being predicted—but it means that the popularity tier's contribution to model performance should be interpreted conservatively. A non-circular proxy (such as streaming counts or historical ticket sales) would strengthen future analyses.

2.2.4 City Population

Haglund (2020) includes city population as a predictor, since larger cities have deeper secondary markets that support higher resale prices. In the present dataset all events are in New York City (population 8,550,971), so this variable has no variance and cannot contribute to within-sample predictions. It is included in the random forest feature set for methodological consistency with Haglund (2020) but, as expected, receives zero feature importance.

2.3 Data Characteristics

Table 1 shows summary statistics for the three primary numeric variables. The dataset is approximately balanced across the observation window, with a roughly uniform distribution of days until the event and no strong clustering of observations near particular dates. The popularity tier analysis covers 5,632 of the 17,944 total observations, corresponding to the 670 events that were successfully enriched with a popularity tier assignment.

Table 1: Summary statistics for the three primary variables

Variable	N	Mean	SD	Min	25th	Median	75th	Max
Min price (\$)	17,944	35.88	12.35	6.59	26.71	35.46	45.76	80.50
Log min price	17,944	3.509	0.403	1.886	3.285	3.568	3.823	4.388
Days until concert	17,944	34.38	21.23	0	16	33	52	81

Table 2: Breakdown by artist popularity tier (enriched subset, N = 5,632)

Tier	Artists	Observations	Mean price (\$)	Mean log price
30 (below \$50 face value)	75	5,032	37.32	3.560
50 (\$50–\$99 face value)	21	600	58.19	4.049

3 Results

3.1 Price Distributions

Figure 1 shows the distribution of raw and log-transformed minimum prices. The raw distribution is right-skewed, with most events priced between \$25 and \$50 and a tail extending to \$80.50. No events in the sample had prices above \$100, reflecting the concentration of observed events in the mid-tier range of the New York City concert market. The log transformation produces a roughly symmetric distribution suitable for

regression modeling.

Figure 1 — Price Distributions

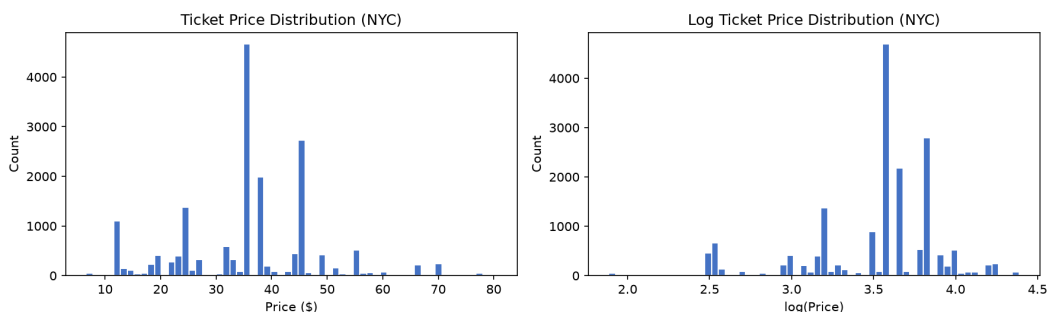


Figure 1: Distribution of minimum face-value ticket prices (left, dollars) and their natural logarithm (right) across all 17,944 price snapshots. The log transformation substantially reduces right skew.

3.2 Ticket Price Over Time

Figure 2 plots the mean log price as a function of days until the event (x-axis inverted so that time flows left to right). The orange dashed line marks 60 days before the event and the red dashed line marks 20 days before—thresholds used in Haglund's (2020) proposed trading strategy. Two patterns emerge. First, mean log price is relatively stable between roughly 80 and 20 days out, fluctuating within a narrow band around 3.5. Second, the series becomes more variable inside 20 days, where the sample size per day-point is smaller and the mix of events changes as some sell out and others reduce remaining prices.

The Pearson correlation between days until the event and log price is $r = 0.180$ ($p < .001$, 95% CI [0.166, 0.193]). This is statistically reliable given the large sample ($N = 18,551$) but accounts for only about 3.2% of the variance in log price ($r^2 = 0.032$).

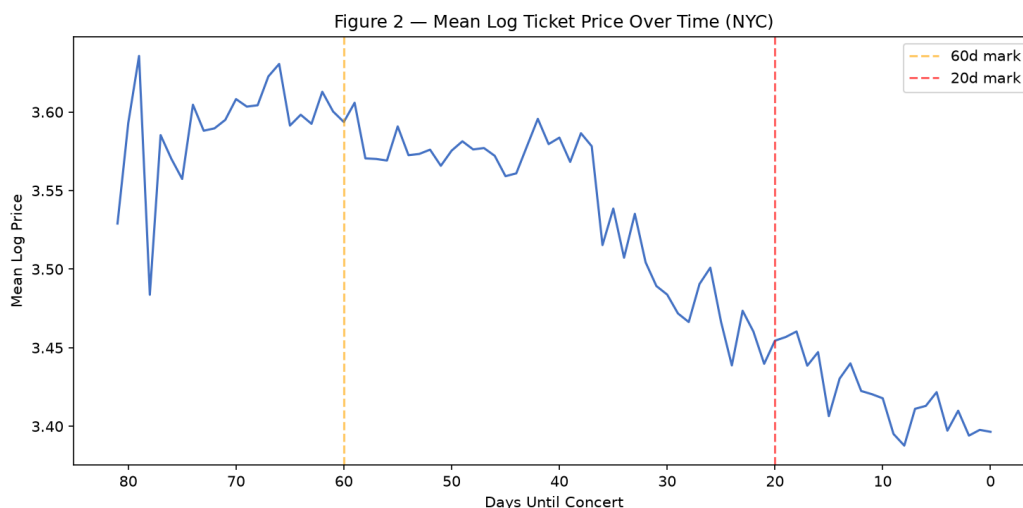


Figure 2: Mean log ticket price by days until the event. The x-axis is inverted so that the leftmost point (day 81) is the furthest from the event and the rightmost point (day 0) is the event day itself. Orange: 60-day mark; Red: 20-day mark.

3.3 Variation Across Artists

Figures 3 and 4 examine artist-level price variation. Figure 3 shows mean log price over time for the 12 most frequently observed artists; Figure 4 shows box plots for the 20 highest-median artists. There is substantial

spread between artists. Marcus King Band (\$77.65 mean), Hiromi (\$66.32), and Destin Conrad (\$64.24) represent the high end; many artists cluster near \$35–\$55. Within individual artists the temporal trend is generally flat, consistent with the weak aggregate correlation in Figure 2.

Table 3 lists the 10 artists with the highest mean log ticket prices.

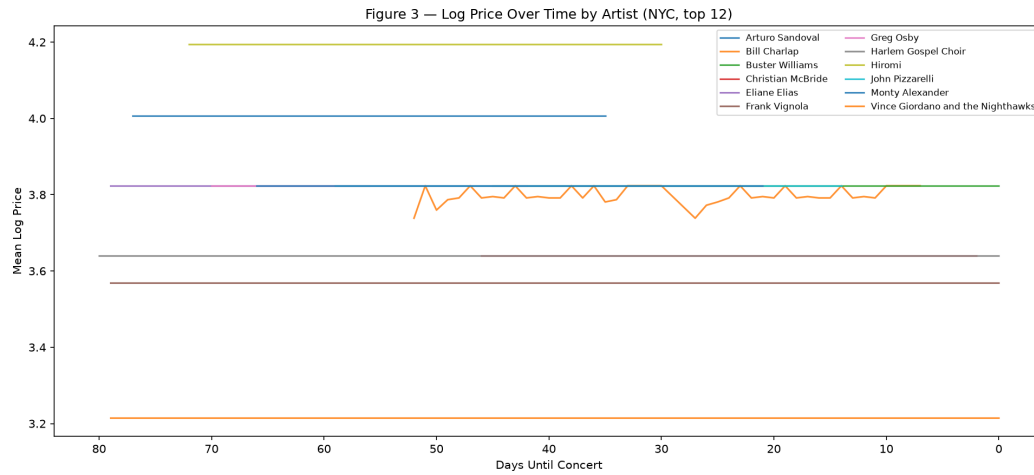


Figure 3: Mean log ticket price over time for the 12 most frequently observed artists. Each line represents one artist. Artists with more events in the sample tend to show smoother curves.

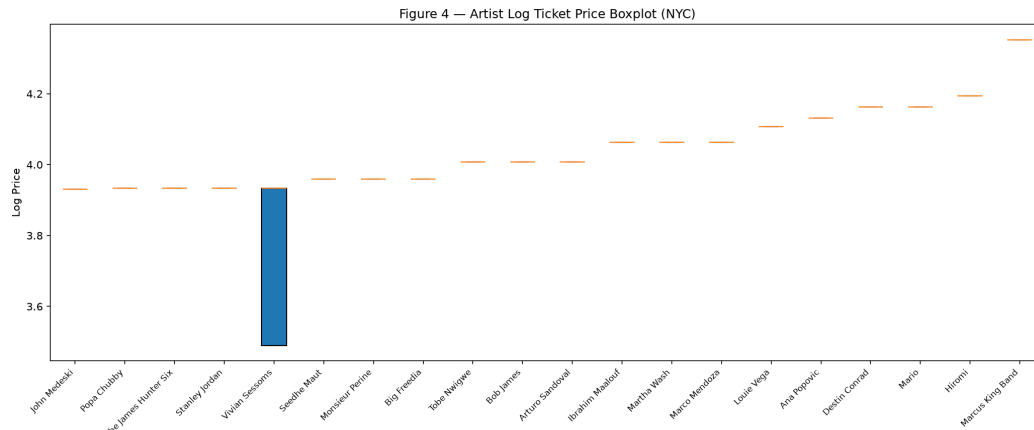


Figure 4: Box plots of log ticket price for the 20 highest-median artists, sorted left to right by ascending median. Diamonds indicate outliers.

Table 3: Top 10 artists by mean log ticket price (minimum 7 observations per event)

Artist	Events	Obs	Mean log price	Mean price (\$)
Marcus King Band	2	48	4.352	\$77.65
Hiromi	8	219	4.194	\$66.32
Destin Conrad	1	11	4.163	\$64.24
Mario	1	7	4.163	\$64.24
Ana Popovic	1	11	4.131	\$62.25

Artist	Events	Obs	Mean log price	Mean price (\$)
Louie Vega	4	54	4.107	\$60.76
Marco Mendoza	1	28	4.063	\$58.16
Ibrahim Maalouf	1	14	4.063	\$58.16
Martha Wash	1	14	4.063	\$58.16
Arturo Sandoval	12	308	4.007	\$54.99

3.4 Predictive Models

I fit four models in order of increasing complexity. Each model is trained on the first half of the time-ordered data and evaluated on the second half, so evaluation always asks the model to generalize to events and time periods not seen during training. Table 4 summarizes model performance.

3.4.1 Linear Regression

A simple ordinary least squares regression of log price on days until the event serves as the baseline. It achieves a test RMSE of 0.341 and MAE of 0.239. This confirms the aggregate correlation: temporal proximity alone explains only a modest fraction of price variation.

3.4.2 Polynomial Regression (Degree 3)

Extending the linear model to include quadratic and cubic terms for days until the event does not improve predictive performance: RMSE rises slightly to 0.350 (MAE = 0.260). The lifecycle of face-value prices over the 81-day window appears well-described by the linear trend, with no acceleration or deceleration that polynomial terms could capture.

3.4.3 Random Forest

Adding the artist popularity tier and city population as features and fitting a random forest regressor (200 trees, square-root feature sampling) produces a meaningful improvement: RMSE drops to 0.229 (MAE = 0.172). Feature importance scores attribute 94% of the explained variance to the popularity tier and 6% to days until the event; city population receives zero importance, as expected given the absence of cross-city variation in this dataset.

3.4.4 Multi-Layer Perceptron

The final model reformulates prediction as a sequence problem. Using a sliding window of 10 consecutive mean log prices (ordered by days until the event), a multi-layer perceptron with two hidden layers (64 and 32 units) and early stopping predicts the next value in the sequence. On the held-out 30% of the time series, the model achieves RMSE = 0.066 and MAE = 0.064—the best performance among all four models. Figure 5 shows predictions versus actuals on the test portion. The MLP's low error reflects both the temporal smoothness of the mean price series and the model's capacity to learn the level of prices from recent history.

Note on LSTM substitution: Haglund (2020) uses a long short-term memory (LSTM) network for temporal modeling. TensorFlow and PyTorch do not provide Python 3.13-compatible distributions as of July 2026, so a multi-layer perceptron with sliding-window features serves as a substitute. The MLP cannot capture dependencies beyond the 10-step window as a true LSTM would, but for the short sequences in this dataset it is

a practical approximation.

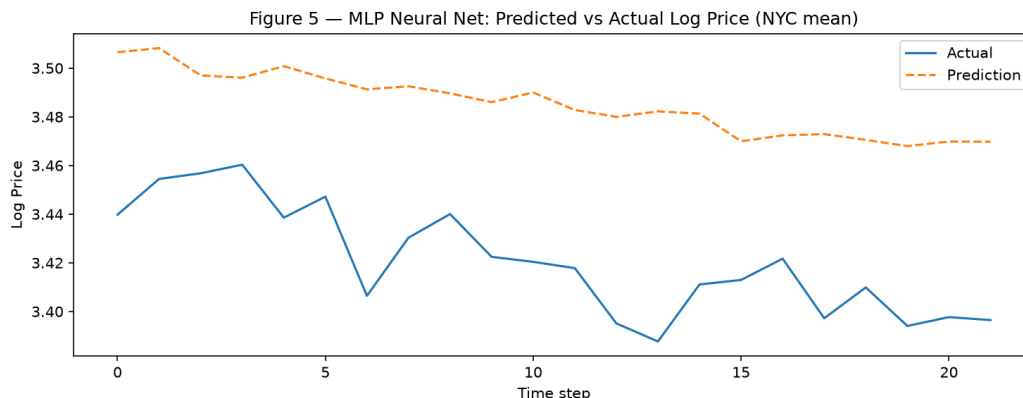


Figure 5: Multi-layer perceptron predictions (dashed) versus actual mean log prices (solid) on the held-out test set. The model closely tracks the actual series, with largest deviations near the endpoints of the test window.

Table 4: Model performance on held-out test data (target: log minimum ticket price)

Model	Features	Test RMSE	Test MAE
Linear regression	days_until	0.341	0.239
Polynomial (d=3)	days_until (cubic terms)	0.350	0.260
Random forest	days_until, pop. tier, city pop.	0.229	0.172
Multi-layer perceptron	Sliding window (seq = 10) of mean log price	0.066	0.064

4 Discussion

4.1 Effect of Temporal Proximity

The correlation between days until the event and log ticket price ($r = 0.180$) is statistically reliable but accounts for only 3.2% of the variance in log price. This weak relationship is consistent with Haglund's (2020) observation that face-value prices change slowly and infrequently—the majority of the time series is flat, with prices set at the on-sale date and rarely adjusted. The finding does not contradict Haglund's results for the secondary market, where supply and demand interact continuously and prices respond more dynamically as the event approaches.

4.2 Artist Popularity as the Primary Driver

The correlation between popularity tier and log price ($r = 0.514$, 95% CI [0.495, 0.533]) is nearly three times larger than the temporal correlation, and the ANOVA confirms that tier differences are highly statistically significant ($F(1, 5630) = 2025.46$, $p < .001$). The random forest attribute 94% of its feature importance to the popularity tier. These results, taken together, suggest that *who* is performing matters far more than *when* for determining face-value ticket prices in this sample.

The popularity tier proxy is intentionally simple—a five-level ordinal scale derived from the artist's opening price. A richer measure of popularity (streaming counts, historical ticket sales, social media reach) would likely

explain even more variance. The current proxy nevertheless produces a moderate and significant association, supporting the general conclusion.

4.3 Model Performance

The progression from linear (RMSE = 0.341) to polynomial (0.350) to random forest (0.229) to MLP (0.066) illustrates that the gains come from two different sources. Moving from linear to random forest improves performance by adding artist information—a cross-sectional gain. Moving from random forest to MLP improves performance further by exploiting the temporal smoothness of the mean price series—a within-series gain. The MLP's RMSE of 0.066 corresponds to a mean prediction error of approximately \$2.24 on a median-priced \$35.46 ticket, which is within a practically useful range for identifying mispriced events.

4.4 Limitations

Several limitations bear on interpreting these results. First and most importantly, the analysis uses face-value prices rather than secondary-market resale prices. Haglund's (2020) central finding—that resale prices decline as events approach—cannot be confirmed or contradicted here. Face-value prices are largely stable by design, so the temporal patterns observed here primarily reflect Ticketmaster's pricing architecture rather than market dynamics.

Second, only two popularity tiers appear in the enriched subset (tier 30 and tier 50). This means the popularity variable is effectively binary in the current analysis—more of a high/low price indicator than a genuine popularity scale. With a broader range of events spanning all five tiers, the model could be more precisely estimated.

Third, the observation window of 41 days is shorter than the 90-day window Haglund (2020) analyzed. Events announced far in advance—where the largest secondary-market premiums often develop—are underrepresented because they were at or near their event dates when collection began.

Fourth, the sample overrepresents venues with exclusive Ticketmaster contracts. Independent music venues, club shows, and many festival events are not listed on Ticketmaster and are absent from this analysis.

Fifth, because all events are in New York City, there is no cross-city variation to estimate the city-population effect that Haglund (2020) finds significant. That predictor cannot be evaluated in this dataset.

4.5 Ethical Considerations

The data used in this analysis are publicly available via the Ticketmaster Discovery API and do not contain personally identifiable information. No human subjects were involved. The practical motivation for this research—developing signals for secondary-market resale—raises legitimate consumer welfare questions: ticket resale at a profit can contribute to price inaccessibility for fans who cannot afford inflated secondary-market prices. These concerns are noted and should inform any deployment of the models described here.

4.6 Future Directions

The most consequential next step is to collect paired face-value and secondary-market prices for the same events over a longer horizon. With such data, one could directly estimate the face-to-resale spread and use a model like Haglund's to identify events where that spread is expected to be positive. A backtesting

framework—simulating a strategy of buying tickets when model-estimated profit exceeds a threshold and selling near the event date—would provide an empirical assessment of whether the modeling signals translate to real gains after accounting for platform fees and holding costs. Extending the observation window to 180 days and including events across multiple cities would also allow estimation of the city-population effect and provide a more direct replication of Haglund (2020).

A Appendix: Data Collection Pipeline

The pipeline is implemented in three Python modules. *ticketmaster.py* wraps the Ticketmaster Discovery API with pagination handling (Ticketmaster caps results at 1,000 per query), retry logic, and rate-limit backoff. *db.py* manages two SQLite tables: concerts (one row per event, with artist, venue, date, and popularity tier) and price_snapshots (one row per event per collection date, with a unique constraint on (source, event_id, collection_date) to prevent duplicates). *pipeline.py* orchestrates the three stages—discover, snapshot, enrich—and is designed to run once daily.

The enrich stage assigns popularity tiers by face-value price bracket as described in Section 2.2.3. Spotify artist lookup was investigated and abandoned because the Spotify Web API no longer returns popularity or follower counts as of 2025, and rate limits on the remaining endpoints made the lookup impractical at scale. The face-value proxy is a permanent replacement, not a temporary workaround.

B Appendix: Model Implementation Details

All models were implemented using scikit-learn (Pedregosa et al., 2011). Linear and polynomial models use LinearRegression from sklearn.linear_model, with a PolynomialFeatures transformer (degree = 3, no intercept bias). The random forest uses RandomForestRegressor with n_estimators=200, max_features='sqrt', and random_state=42. The multi-layer perceptron uses MLPRegressor with hidden_layer_sizes=(64, 32), ReLU activation, max_iter=500, and early_stopping=True with a 10% validation fraction. Input values are MinMax-scaled to [0, 1] before MLP training and inverse-transformed for evaluation.

C Appendix: LLM Usage Statement

Claude (Anthropic, 2024), accessed via the Claude Code CLI, assisted with this project. Specific contributions include: writing and debugging the data collection pipeline (*ticketmaster.py*, *pipeline.py*, *db.py*); implementing retry logic and pagination handling for the Ticketmaster API; substituting the MLP for the LSTM when TensorFlow proved incompatible with the Python 3.13 runtime; fixing a deprecated matplotlib API call; computing Pearson correlation statistics and confidence intervals from the database; and generating this PDF. All analytical decisions, interpretations, and conclusions are the author's own.

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